

## SHANNON, VA, USA

WMO: 724033

Lat: 38.267N

Lon: 77.449W

Elev: 26

StdP: 101.01

Time Zone: -5.00 (NAE)

Period: 94-19

WBAN: 03706

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |       |     |      | Coldest Month WS/MCDB |      |     |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-------|-----|------|-----------------------|------|-----|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99%   |     |      | 0.4%                  |      | 1%  |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP    | HR  | MCDB | WS                    | MCDB | WS  | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g)   | (h) | (i)  | (j)                   | (k)  | (l) | (m)  | (n)                   | (o)  | (p)   |
| 1             | -9.4       | -7.4 | -18.0                         | 0.8 | -6.3 | -15.5 | 1.0 | -4.5 | 10.3                  | 7.5  | 9.1 | 6.5  | 1.5                   | 330  | 0.415 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 11.1                   | 35.2            | 25.5 | 33.8 | 24.8 | 32.5 | 24.3 | 26.9                | 32.4 | 26.1 | 31.5 | 25.4 | 30.3 | 3.1                  | 210  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 25.7                            | 21.0 | 29.0 | 24.8 | 19.9 | 28.3 | 24.0 | 18.9 | 27.7 | 84.4          | 32.0 | 81.0 | 31.3 | 77.7 | 30.6 | 30.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |     | Extreme Annual Temperature |       |                    |     | n-Year Return Period Values of Extreme Temperature |       |            |       |            |       |            |       |      |     |
|-------------------|------|-----|----------------------------|-------|--------------------|-----|----------------------------------------------------|-------|------------|-------|------------|-------|------------|-------|------|-----|
|                   |      |     | Mean                       |       | Standard Deviation |     | n=5 years                                          |       | n=10 years |       | n=20 years |       | n=50 years |       |      |     |
| 1%                | 2.5% | 5%  | Min                        | Max   | Min                | Max | Min                                                | Max   | Min        | Max   | Min        | Max   | Min        | Max   |      |     |
| (a)               | (b)  | (c) | (e)                        | (f)   | (g)                | (h) | (i)                                                | (j)   | (k)        | (l)   | (m)        | (n)   | (o)        | (p)   |      |     |
| (4) 8.4           | 7.3  | 6.1 | DB                         | -14.2 | 37.2               | 3.5 | 1.6                                                | -16.7 | 38.3       | -18.7 | 39.2       | -20.7 | 40.1       | -23.2 | 41.2 | (4) |
| (5)               |      |     | WB                         | -14.6 | 27.9               | 3.4 | 1.1                                                | -17.0 | 28.6       | -19.0 | 29.2       | -20.9 | 29.8       | -23.3 | 30.6 | (5) |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |      |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |      |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 14.6        | 2.8   | 4.3   | 8.4   | 14.2  | 19.2  | 23.7  | 26.0  | 25.3  | 21.5  | 15.1  | 9.0   | 4.9   | (6)  |  |
|                                                                        | DBStd       | 9.36        | 5.99  | 5.44  | 5.53  | 4.99  | 4.35  | 3.36  | 2.72  | 2.74  | 3.63  | 4.60  | 4.77  | 5.01  | (7)  |  |
|                                                                        | HDD10.0     | 778         | 236   | 173   | 99    | 15    | 0     | 0     | 0     | 0     | 0     | 8     | 73    | 174   | (8)  |  |
|                                                                        | HDD18.3     | 2208        | 481   | 393   | 311   | 143   | 43    | 4     | 0     | 0     | 13    | 122   | 280   | 417   | (9)  |  |
|                                                                        | CDD10.0     | 2455        | 13    | 14    | 49    | 141   | 286   | 411   | 496   | 475   | 343   | 167   | 44    | 15    | (10) |  |
|                                                                        | CDD18.3     | 845         | 1     | 1     | 3     | 20    | 71    | 164   | 238   | 217   | 107   | 22    | 1     | 1     | (11) |  |
|                                                                        | CDH23.3     | 8202        | 2     | 3     | 46    | 260   | 708   | 1584  | 2432  | 2077  | 877   | 197   | 14    | 3     | (12) |  |
|                                                                        | CDH26.7     | 3433        | 0     | 0     | 8     | 83    | 249   | 662   | 1139  | 928   | 311   | 51    | 1     | 0     | (13) |  |
| (14) Wind                                                              | WSAvg       | 2.2         | 2.5   | 2.6   | 2.9   | 2.8   | 2.2   | 2.0   | 1.7   | 1.6   | 1.8   | 1.9   | 2.1   | 2.1   | (14) |  |
| (15) Precipitation                                                     | PrecAvg     | 1047        | 81    | 74    | 97    | 77    | 99    | 85    | 104   | 89    | 97    | 89    | 84    | 85    | (15) |  |
|                                                                        | PrecMax     | 1516        | 192   | 162   | 243   | 177   | 219   | 274   | 297   | 254   | 278   | 202   | 184   | 180   | (16) |  |
|                                                                        | PrecMin     | 706         | 9     | 7     | 25    | 10    | 16    | 18    | 20    | 13    | 13    | 4     | 5     | 18    | (17) |  |
|                                                                        | PrecStd     | 206         | 38    | 41    | 47    | 36    | 49    | 50    | 66    | 58    | 63    | 50    | 48    | 47    | (18) |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 22.3  | 22.9  | 27.7  | 32.2  | 33.1  | 35.9  | 37.5  | 37.1  | 34.2  | 30.9  | 25.6  | 22.7  | (19) |  |
|                                                                        |             | MCWB        | 16.1  | 15.8  | 17.8  | 20.7  | 23.2  | 25.7  | 26.2  | 25.4  | 24.1  | 22.8  | 17.9  | 16.5  | (20) |  |
|                                                                        | 2%          | DB          | 18.7  | 19.1  | 24.0  | 29.0  | 31.5  | 33.7  | 35.4  | 34.9  | 32.3  | 27.8  | 22.4  | 18.8  | (21) |  |
|                                                                        |             | MCWB        | 15.3  | 13.5  | 15.5  | 18.9  | 22.6  | 24.6  | 25.8  | 25.5  | 23.7  | 20.7  | 16.3  | 15.5  | (22) |  |
|                                                                        | 5%          | DB          | 16.2  | 16.3  | 21.3  | 26.4  | 29.7  | 32.3  | 33.9  | 33.0  | 30.2  | 25.4  | 20.0  | 16.1  | (23) |  |
|                                                                        |             | MCWB        | 12.4  | 11.1  | 14.3  | 17.8  | 21.7  | 24.2  | 25.1  | 24.8  | 22.7  | 19.5  | 15.1  | 12.5  | (24) |  |
|                                                                        | 10%         | DB          | 12.4  | 13.1  | 18.2  | 23.7  | 27.6  | 30.9  | 32.4  | 31.7  | 27.9  | 22.9  | 17.6  | 13.0  | (25) |  |
|                                                                        |             | MCWB        | 8.8   | 8.6   | 12.4  | 16.2  | 20.5  | 23.5  | 24.7  | 24.1  | 21.6  | 18.4  | 13.5  | 9.7   | (26) |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 18.2  | 17.5  | 18.8  | 22.7  | 25.4  | 27.3  | 28.4  | 27.9  | 25.9  | 24.3  | 19.5  | 18.9  | (27) |  |
|                                                                        |             | MCDB        | 19.8  | 21.4  | 25.2  | 28.5  | 30.4  | 32.9  | 34.0  | 33.1  | 30.3  | 27.6  | 23.0  | 20.6  | (28) |  |
|                                                                        | 2%          | WB          | 15.5  | 14.9  | 17.0  | 20.5  | 24.0  | 26.0  | 27.1  | 26.7  | 25.0  | 22.8  | 17.9  | 16.1  | (29) |  |
|                                                                        |             | MCDB        | 18.6  | 18.1  | 22.3  | 26.8  | 29.0  | 31.1  | 32.5  | 32.3  | 29.6  | 25.9  | 20.7  | 18.4  | (30) |  |
|                                                                        | 5%          | WB          | 12.7  | 11.8  | 15.3  | 18.9  | 22.8  | 25.1  | 26.3  | 25.8  | 24.1  | 21.1  | 16.3  | 13.2  | (31) |  |
|                                                                        |             | MCDB        | 15.8  | 14.7  | 20.1  | 24.6  | 27.6  | 30.5  | 31.8  | 30.9  | 28.0  | 24.4  | 19.3  | 15.5  | (32) |  |
|                                                                        | 10%         | WB          | 9.1   | 9.2   | 13.2  | 17.3  | 21.7  | 24.3  | 25.5  | 25.1  | 23.1  | 19.3  | 14.0  | 10.2  | (33) |  |
|                                                                        |             | MCDB        | 11.8  | 13.0  | 17.6  | 22.2  | 26.2  | 29.0  | 30.5  | 29.6  | 26.1  | 22.1  | 16.8  | 12.5  | (34) |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 10.3  | 11.3  | 12.1  | 13.0  | 12.0  | 11.3  | 11.1  | 11.2  | 11.3  | 12.2  | 12.3  | 10.6  | (35) |  |
|                                                                        |             | MCDBR       | 14.4  | 15.3  | 16.4  | 16.5  | 14.5  | 13.1  | 12.8  | 13.1  | 13.7  | 14.8  | 15.6  | 13.8  | (36) |  |
|                                                                        | 5% WB       | MCWBR       | 11.0  | 10.5  | 9.8   | 8.1   | 7.0   | 5.8   | 4.8   | 5.1   | 5.9   | 8.2   | 9.9   | 10.4  | (37) |  |
|                                                                        |             | MCWBR       | 13.3  | 13.4  | 14.1  | 14.1  | 12.3  | 11.7  | 11.5  | 11.4  | 10.8  | 11.9  | 12.2  | 12.1  | (38) |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.304 | 0.318 | 0.354 | 0.395 | 0.434 | 0.488 | 0.508 | 0.509 | 0.416 | 0.358 | 0.323 | 0.308 | (40) |  |
|                                                                        |             | taud        | 2.509 | 2.451 | 2.403 | 2.304 | 2.224 | 2.113 | 2.082 | 2.059 | 2.300 | 2.464 | 2.519 | 2.534 | (41) |  |
|                                                                        | Edn at Noon |             | 882   | 911   | 907   | 883   | 851   | 803   | 782   | 770   | 829   | 854   | 850   | 851   | (42) |  |
|                                                                        |             | Edh at Noon | 78    | 94    | 109   | 127   | 141   | 157   | 161   | 161   | 119   | 92    | 77    | 72    | (43) |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 2.19  | 2.99  | 3.97  | 4.96  | 5.62  | 6.16  | 5.97  | 5.36  | 4.38  | 3.36  | 2.47  | 1.93  | (44) |  |
|                                                                        | RadStd      |             | 0.19  | 0.33  | 0.29  | 0.37  | 0.57  | 0.45  | 0.31  | 0.26  | 0.50  | 0.38  | 0.20  | 0.18  | (45) |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |      |  |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|------|--|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |      |  |
|      |                         | (d)   | (e)     | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |      |  |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | +1.05 | N/A         | N/A     | N/A     | N/A     | (46) |  |
| (47) | Regional (74 neighbors) | +0.43 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +138    | +84     | (47) |  |

Nomenclature: See separate page